Time-Series Forecasting of Appliance Energy Use:
A Comparative Evaluation of Benchmark, SARIMAX, Machine-Learning and Foundation-Model Approaches

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1. Introduction and Problem Definition

This report develops and critically compares time-series forecasting models for hourly appliance electricity consumption, using the Appliances Energy Prediction dataset collected from a low-energy residential building in Belgium (Candanedo, Feldheim and Deramaix, 2017). The dataset combines indoor temperature and humidity readings from a wireless sensor network, weather observations from a nearby airport station, and the target variable itself, recorded at 10-minute resolution between mid-January and the end of May 2016.

The forecasting target is hourly appliance energy consumption in Wh, obtained by summing the 10-minute readings within each hour rather than averaging them, since Wh is a quantity of energy rather than an instantaneous rate; summation preserves the physical meaning of the target after resampling. The forecast horizon is 24 hours, evaluated continuously across a 336-hour (14-day) held-out test period at the end of the series, with all preceding data used for training. This split respects the temporal ordering of the data and avoids leakage from future observations into model fitting.

Model accuracy is compared using four complementary metrics: mean absolute error (MAE) and root mean squared error (RMSE) in the original Wh units, mean absolute scaled error (MASE), and bias (mean signed error). MASE scales absolute error against the in-sample error of a naive benchmark, giving a metric that is comparable across series and interpretable relative to a trivial forecast, with values below 1 indicating that a model beats the reference method (Hyndman and Koehler, 2006). Bias is reported separately because a model can have a competitive average error while still being systematically skewed in one direction, which matters for a deployment context such as smart-home energy management.

2. Data Preparation and Exploratory Analysis

The raw dataset contained no missing values across any column at 10-minute resolution, so no imputation was required. Data were resampled to hourly frequency as described above, and initial plots were produced to characterise the series before modelling.

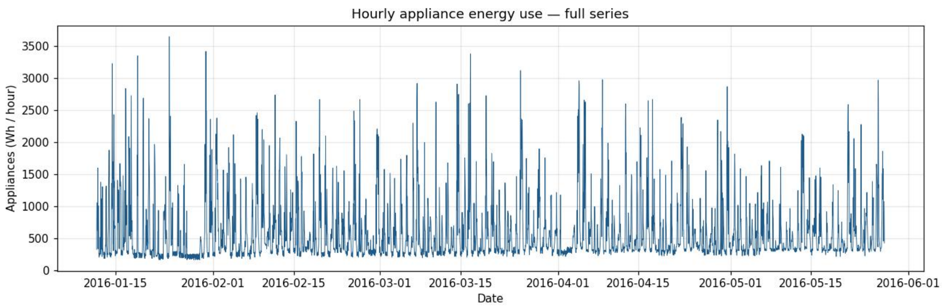

Figure 1. Hourly appliance energy use across the full series (mid-January to end of May 2016).

The full series (Figure 1) shows hourly appliance use ranging from lows of around 200 Wh overnight to spikes exceeding 3,000 Wh, with a strongly right-skewed distribution: most hours cluster below 500 Wh, with a long tail driven by occasional high-demand events. Because this skew and the presence of extreme spikes are the main sources of forecasting difficulty, they are revisited throughout the results and discussion.

Seasonality was assessed by profiling average demand by hour-of-day and day-of-week.

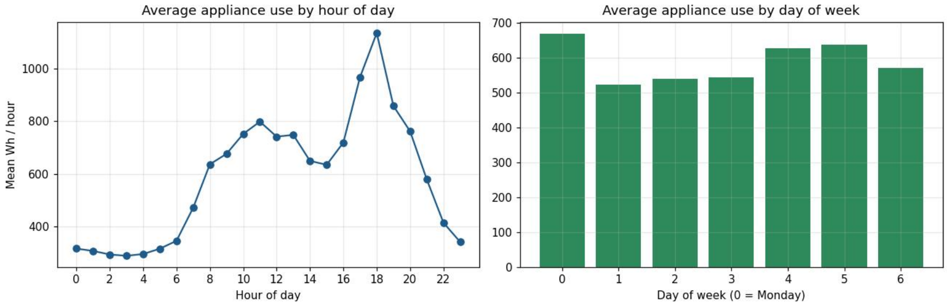

Figure 2. Average appliance use by hour of day (left) and day of week (right).

The hourly profile shows a clear evening peak around 18:00 (approx. 1,140 Wh) and an overnight trough between 02:00 and 04:00 (approx. 290 Wh), consistent with occupancy-driven appliance use. The weekly profile shows Monday as the highest-demand day (approx. 670 Wh average) with a mid-week dip and a smaller weekend rise; this weekly pattern is weaker than the daily one but genuinely present, which becomes relevant when comparing the daily and weekly seasonal-naive benchmarks in Section 5.

Autocorrelation was examined via the ACF and PACF (Figure 4, Section 3.2), which shows strong short-lag autocorrelation decaying quickly, plus a secondary, weaker peak around lag-24, confirming that a daily seasonal component sits on top of short-range persistence rather than dominating it. This directly informed the SARIMAX specification below.

3. Methodology

3.1 Benchmark Models

Five standard benchmark forecasts were produced as a baseline against which all subsequent models are judged (Hyndman and Athanasopoulos, 2021): the historical mean; a naive (persistence) forecast; a daily seasonal-naive forecast (repeats the value from 24 hours earlier); a weekly seasonal-naive forecast (repeats the value from 168 hours earlier); and a drift forecast (naive plus a linear trend extrapolated from the training period).

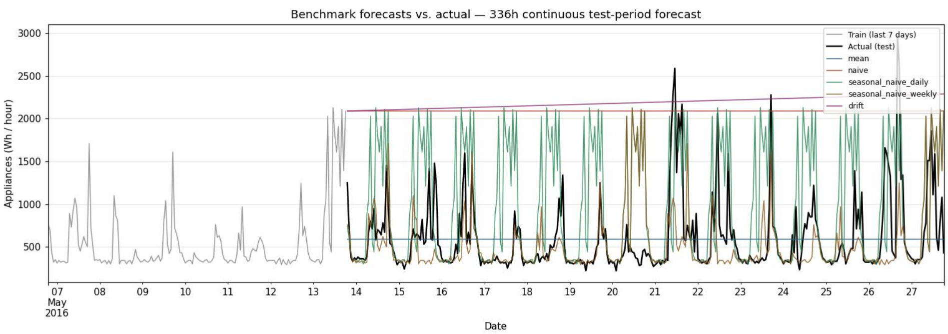

Figure 3. Benchmark forecasts against the actual test-period series.

Figure 3 plots all five benchmark forecasts against the actual test-period series. The naive and drift forecasts are visibly the weakest: because appliance use resets to a low baseline every few hours, a flat persistence forecast or a monotonically increasing drift forecast diverges quickly from the true, cyclical pattern. Both seasonal-naive variants track the broad shape of the series far more closely, since they reintroduce a cyclical structure rather than assuming persistence or a trend.

3.2 SARIMAX

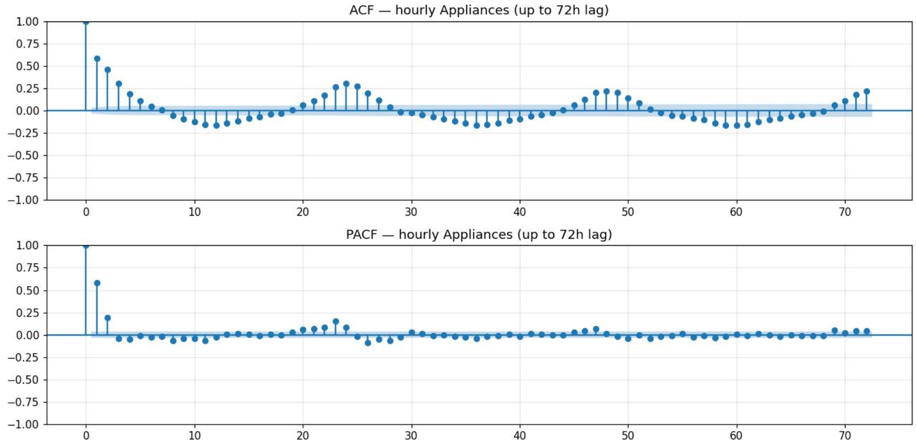

Figure 4. ACF and PACF of hourly appliance use, used to inform the SARIMAX order.

The SARIMAX order was selected using a grid search over the non-seasonal (p, d, q) and seasonal (P, D, Q, 24) components, minimising AIC subject to model convergence, informed by the ACF/PACF diagnostics in Figure 4. The PACF cuts off sharply after lag 2-3, supporting a low, parsimonious autoregressive order, while the seasonal grid search confirmed that adding a seasonal moving-average term at the daily period materially improved fit: AIC fell from 43,529.9 with no seasonal AR/MA structure to 43,050.7 for a seasonal order of (0,1,1,24), the strongest seasonal specification found. The final model combined this seasonal structure with a parsimonious non-seasonal order and exogenous regressors, consistent with the recommendation of Box et al. (2015) to derive SARIMAX specifications from autocorrelation diagnostics rather than from defaults.

3.3 Sensor, Weather and Time-Based Covariates

To support the feature-based model, a covariate set was engineered from the raw sensor and weather channels plus derived time features: lagged appliance-use values (1, 2, 3, 24, 48 and 168 hours), rolling mean/standard-deviation windows (3, 6, 12 and 24 hours), cyclical encodings of hour-of-day and day-of-week (sine/cosine pairs), and raw indoor temperature/humidity and outdoor weather channels. A simple correlation screen showed the strongest linear relationships were with the most recent lags (lag-1 correlation approx. 0.59, roll-mean-3 approx. 0.55) and the cyclical hour encodings, with outdoor humidity the strongest weather-derived signal (approx. -0.20).

3.4 Feature-Based Machine-Learning Model

An XGBoost regressor (Chen and Guestrin, 2016) was trained on the full covariate set described above, with hyperparameters tuned to balance fit against overfitting risk.

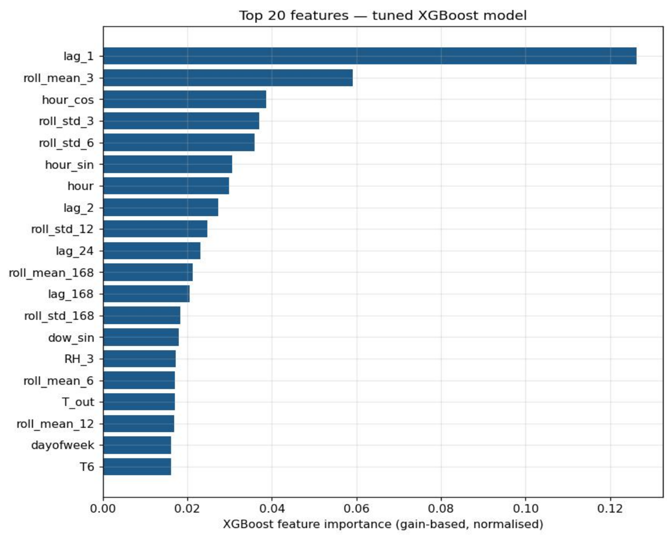

Figure 5. Top 20 features by gain-based importance, tuned XGBoost model.

Gain-based feature importance (Figure 5) shows lag-1 dominating (importance approx. 0.13, roughly double the next feature), followed by roll-mean-3 and the cyclical hour encodings. Notably, the model weights the hour-of-day features considerably more heavily than the simple linear correlation screen in Section 3.3 would suggest, implying it is capturing non-linear time-of-day interactions that correlation alone misses.

3.5 Time-Series Foundation Model

Chronos-Bolt (Ansari et al., 2024), a pretrained transformer-based probabilistic forecasting model, was applied in zero-shot mode, i.e. without any training or fine-tuning on this dataset, to test whether a general-purpose time-series foundation model can compete with classical and feature-based approaches trained specifically on this series. Related foundation models such as TimesFM (Das et al., 2023) follow a similar decoder-only pretraining paradigm; Chronos was selected here for its established zero-shot performance on unseen series.

4. Evaluation Framework

All models were evaluated on the identical 336-hour held-out test period using MAE, RMSE, MASE and Bias (Section 1), with the strongest benchmark, weekly seasonal-naive (MASE = 0.809), used as the MASE reference point. In addition to aggregate metrics, forecasts were compared visually against the true series (Section 5) and residual diagnostics, including autocorrelation of residuals, were examined to check whether each model had captured the available temporal structure (Section 6).

5. Results

Table 1 summarises all eight models' accuracy across the full test period.

Table 1. Forecast accuracy metrics for all models across the 336-hour test period.

Model	MAE	RMSE	MASE	Bias
XGBoost (tuned)	174.02	309.87	0.562	+1.94
Chronos-Bolt (zero-shot)	201.98	379.42	0.630	-69.06
SARIMAX	227.44	375.28	0.710	+18.78
Seasonal-naive (weekly)	259.29	483.55	0.809	-58.51
Mean	296.44	436.14	0.925	-12.25
Seasonal-naive (daily)	525.24	780.20	1.639	+390.48
Naive	1507.32	1555.33	4.702	+1492.98
Drift	1602.91	1650.71	5.000	+1593.40

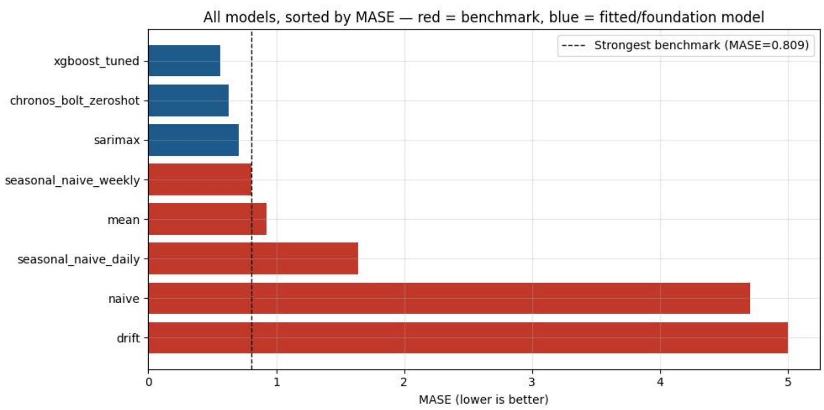

Figure 6. All models ranked by MASE; dashed line marks the strongest benchmark.

Figure 6 ranks all models by MASE. All three fitted/pretrained models (XGBoost, Chronos, SARIMAX) clear the strongest benchmark's MASE of 0.809, confirming that model-based approaches added genuine value over trivial forecasting; naive and drift, in contrast, perform four to five times worse than the strongest benchmark, underscoring the importance of comparing against a properly chosen baseline rather than the weakest possible one.

XGBoost is the best model on every metric (MAE 174.0, RMSE 309.9, MASE 0.562), and its bias (+1.9) is the smallest in magnitude of any model, i.e. it is not just accurate on average but not systematically over- or under-forecasting. Chronos is the second-best model by MAE/MASE despite zero-shot use, but has the largest bias magnitude of the three fitted models (-69.1), meaning it consistently under-forecasts. SARIMAX sits third, ahead of all benchmarks but behind both machine-learning-based approaches.

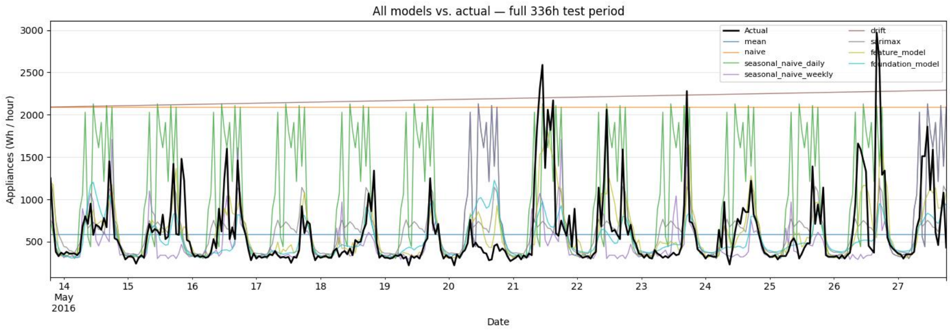

Figure 7. All model forecasts against the actual series across the full 336-hour test period.

Figure 7 overlays every model's forecast against the actual series. It shows visually what the metrics summarise numerically: seasonal-naive-weekly and SARIMAX broadly track the cyclical shape but flatten through demand spikes, while XGBoost and Chronos follow the cyclical pattern more closely and respond somewhat to spike events, though neither reaches the full magnitude of the largest peaks (for example, the 2,970 Wh spike on 26 May, where XGBoost forecast only 546 Wh and Chronos only 652 Wh).

To separate typical-demand accuracy from spike-handling, the test set was split into a high-demand subset (top 10% of actual values, n=35) and a low-demand subset (bottom 50%, n=168).

Table 2. Mean absolute error by demand regime.

Model	Peak-hour MAE (n=35)	Low-demand MAE (n=168)
XGBoost (tuned)	642.4	57.9
Chronos-Bolt (zero-shot)	944.3	74.2
SARIMAX	840.1	146.1
Seasonal-naive (weekly)	943.4	84.8

This breakdown shows that XGBoost's overall advantage is driven mainly by the low-demand regime, where it is roughly 2.5 times more accurate than any other model; at true peaks its advantage narrows substantially, and Chronos is actually less accurate than the weekly seasonal-naive benchmark specifically during peak hours.

6. Discussion and Critical Analysis

The results point to a consistent explanation for why the ranking in Section 5 emerged. Appliance energy use is dominated by short bursts of high demand against a low baseline (Figure 1), so any model that can represent recent local state, via lag and rolling features or via a learned attention mechanism over recent context, has an inherent advantage over models that only encode fixed seasonal or trend structure. This is why XGBoost, with explicit lag-1 through lag-168 features, outperforms SARIMAX, whose seasonal component captures the recurring daily rhythm well (Figure 4) but has no mechanism to react to an unscheduled demand spike beyond its fitted variance.

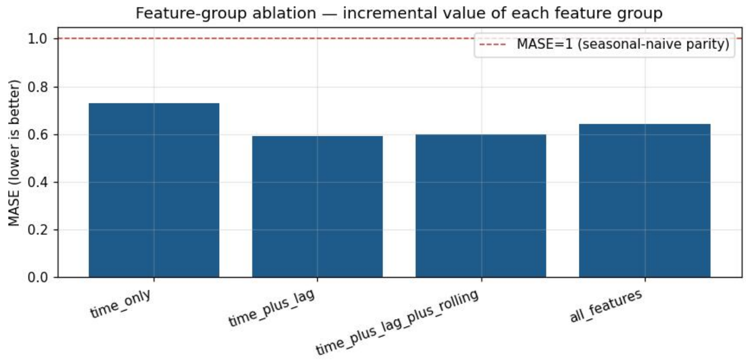

Figure 8. Feature-group ablation showing incremental MASE change as feature groups are added.

The feature-group ablation (Figure 8) supports this explanation directly: adding lag features produced the single largest MASE improvement (time-only MASE 0.730 to time-plus-lag 0.592), while adding rolling-window statistics on top made almost no further difference to MASE (0.592 to 0.600), consistent with rolling features mainly smoothing variance rather than shifting average accuracy. Notably, the complete feature set, including all sensor and weather covariates, has a slightly worse MASE (0.643) than lag-only features, a mild instance of feature dilution: additional weakly relevant covariates add noise for typical points even where they marginally reduce a handful of large errors, which is instead reflected in a lower RMSE for the fuller feature set. This is a genuine trade-off rather than a straightforward improvement, and it argues for feature selection rather than including every available covariate by default.

All models still substantially under-forecast the very largest spikes. This is a structural limitation rather than a tuning failure: the largest test-period events are not explained by any feature available in this dataset (time-of-day, recent lags, or weather), so no model trained only on these inputs can be expected to anticipate them; they most likely reflect ad hoc appliance use, such as a tumble dryer or oven cycle, that is fundamentally unpredictable from the available covariates. The SARIMAX 24-hour forecast interval is also worth flagging as a diagnostic weakness: its 95% confidence interval extends into negative Wh values, which is physically impossible for an energy series, a sign that the model's Gaussian error assumption is a poor fit for this right-skewed target, and that a log-transform or a distribution-aware model would likely produce more realistic uncertainty bounds.

Finally, the XGBoost forecast produced by the underlying pipeline is explicitly labelled a conditional forecast: lag and rolling features for the test period were computed from realised actual values rather than the model's own prior predictions. This is a defensible and common way to evaluate one-step-ahead accuracy in isolation, but it is not equivalent to a genuine 24-hour-ahead forecast that a deployed system would have to produce, since a real deployment would only have its own earlier predictions, or none, available at each step. This distinction is developed further in Section 7.5.

7. Answers to the Set Analysis Questions

7.1 Which benchmark model is strongest, and what does this imply about demand structure?

Weekly seasonal-naive is the strongest benchmark (MASE 0.809), ahead of the mean (0.925), daily seasonal-naive (1.639), naive (4.702) and drift (5.001). That daily seasonal-naive underperforms weekly seasonal-naive is initially counter-intuitive, since the hourly profile (Figure 2) shows a strong daily pattern; the explanation is that day-to-day demand is noisy at the individual-hour level, since a spike at 18:00 one day is not reliably repeated at 18:00 the next, so a naive repeat of "yesterday" carries that noise forward, whereas a repeat of the same hour one week earlier averages over more behavioural variation and happens to align better with weekly routine, given Monday is consistently the highest-demand day. This implies appliance demand has more genuine weekly structure than day-to-day repeatability, a useful and non-obvious finding for any simple forecasting heuristic.

7.2 Does SARIMAX improve on the strongest benchmark?

Yes: SARIMAX (MASE 0.710) improves on weekly seasonal-naive (0.809) but trails both machine-learning approaches. Its residual autocorrelation is close to white noise at most lags, with a mild remaining bump near lag-24, and its forecast plot flattens through demand spikes (Section 6). Together these diagnostics show it captures the smooth daily seasonal cycle adequately but does not fully absorb short-range autocorrelation during volatile periods, and its exogenous regressors do not compensate for the missing local, lag-based information that most drives XGBoost's advantage.

7.3 Does the feature-based model improve with added covariates, and which feature groups help most?

Yes, substantially over a time-only baseline (MASE 0.730 to 0.562 for the tuned, full-feature model), but the improvement is front-loaded: lag features alone deliver almost all of the gain (to 0.592), rolling-window statistics add only marginal further benefit, and the complete covariate set is not uniformly better than lag-only features on every metric (Section 6). Lag and short-window rolling features are clearly the most useful group for this series; the raw sensor and weather covariates contribute comparatively little, most likely because their influence is already captured indirectly through the lag structure and time-of-day encodings.

7.4 Does the foundation model outperform the other approaches, and is any improvement large enough to justify its complexity?

Chronos, used zero-shot with no training on this dataset, outperforms every benchmark and SARIMAX (MASE 0.630 versus SARIMAX's 0.710) but does not outperform the tuned XGBoost model (0.562), and it carries the largest bias magnitude of any fitted model (-69.1). Given that it needed no dataset-specific training to reach this level, the result is a genuinely strong showing for a general-purpose pretrained model, but for this dataset specifically, the additional complexity and compute cost of a transformer-scale foundation model is not justified relative to a comparatively lightweight, purpose-tuned XGBoost model that is both more accurate and easier to interpret and deploy.

7.5 Which variables are genuinely known at forecast origin?

Calendar and time features (hour-of-day, day-of-week) and the target's own lagged history up to the forecast origin are genuinely available in a real deployment, as are indoor sensor readings recorded up to that point. Outdoor weather variables and, crucially, any lag or rolling feature computed from within the forecast horizon itself are not genuinely available at forecast time in a multi-step-ahead setting: using realised test-period values for these, as the XGBoost pipeline does for its reported forecast (Section 6), produces a conditional forecast rather than a true forecast. A deployable version would need either an iterative rollout that feeds the model's own prior predictions back in as lag features, or substitution of real outdoor-weather forecasts, themselves imperfect, in place of realised weather, both of which would be expected to increase error relative to the figures reported here.

7.6 Which model would be recommended for practical smart-home deployment?

Weighing accuracy, interpretability, uncertainty quantification, computational cost and ease of deployment together, XGBoost is the recommended model: it is the most accurate on every metric, its gain-based feature importances (Figure 5) give an interpretable account of what drives its predictions, useful for a homeowner-facing product, and it is cheap to train and run inference with compared to a transformer-scale foundation model, without requiring the specialised infrastructure Chronos does. Its main weakness is uncertainty quantification, which is not native to a point-estimate gradient-boosted model, unlike SARIMAX's confidence intervals or Chronos's native probabilistic output; a production system could address this with a quantile-regression variant of XGBoost. Chronos remains an attractive option where no historical training data yet exists for a new home, given its competitive zero-shot performance.

8. Limitations and Future Work

Several limitations qualify these results. First, the XGBoost evaluation reported here is a conditional, not a true, forecast for the covariates that require future information (Section 7.5); a genuine iterative multi-step evaluation would likely show a smaller, though probably still real, advantage over SARIMAX and Chronos. Second, all models under-forecast the largest demand spikes, which appear driven by ad hoc appliance use not explained by the available covariates; incorporating appliance-level sub-metering or occupancy-sensing data, where available, could plausibly close some of this gap. Third, SARIMAX's Gaussian confidence intervals produce physically implausible negative-Wh bounds, suggesting a log-transformed target or a model with a more appropriate error distribution would give more usable uncertainty estimates. Fourth, Chronos was evaluated purely zero-shot; light fine-tuning on this household's own history, or ensembling Chronos with XGBoost, would be a natural next step, consistent with early evidence that fine-tuned foundation models can close much of the residual gap with specialised models (Ansari et al., 2024). Finally, the dataset spans only around 4.5 months of a single household, which limits how confidently these findings generalise to other households, seasons, or the winter heating period not represented in this data; validating on a second household or a full annual cycle would materially strengthen the conclusions.

9. Conclusion and Model Recommendation

Across benchmark, statistical, machine-learning and foundation-model approaches, a tuned XGBoost model using lag, rolling-window and time-of-day features gave the most accurate and best-calibrated (lowest-bias) forecasts of hourly appliance energy use, comfortably beating both the strongest naive benchmark and a zero-shot time-series foundation model. The zero-shot foundation model's competitive performance without any dataset-specific training is nonetheless a notable finding, and it narrows the practical gap between off-the-shelf and purpose-built forecasting approaches. For smart-home deployment specifically, the recommendation is XGBoost, on accuracy, interpretability and deployment-cost grounds, with the caveats that its evaluated advantage partly reflects a conditional rather than fully causal forecasting setup, and that all models tested share a common blind spot around unscheduled demand spikes that no covariate set in this dataset appears able to resolve.

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